

Problem Set 2

PCC190 - Secure Machine Learning

1. Define the symbol $=$.

- Denotes equality between two quantities that are each already independently defined.
- Asserts that the left-hand expression and the right-hand expression refer to the same object.
- Does not introduce new notation on either side.

2. Define the symbol $\triangleq$.

- Denotes “defined as.”
- Introduces new notation: the symbol on the left is defined, by stipulation, to mean the expression on the right.
- Differs from $=$, which relates two quantities that already exist independently of the statement.

3. Present the formal definition of a learning algorithm given by Mitchell (1997).

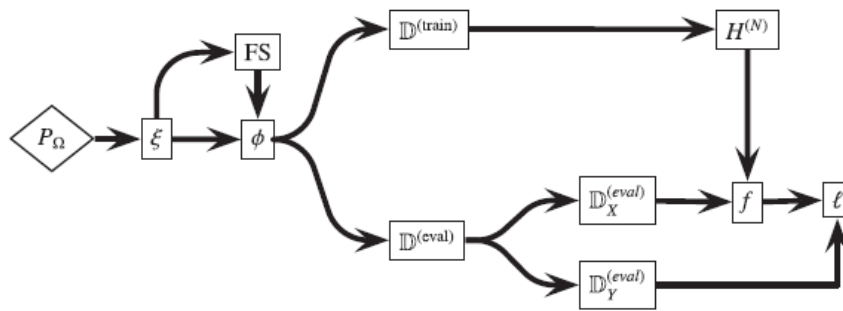
- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .
- Three ingredients: a class of tasks T , a performance measure P , and a source of experience E .
- A program is said to learn when its performance at tasks in T , as measured by P , improves as a function of experience E .
- The definition says nothing about how the improvement is achieved internally.

4. Explain the learning framework depicted in Figure 2.1(a).

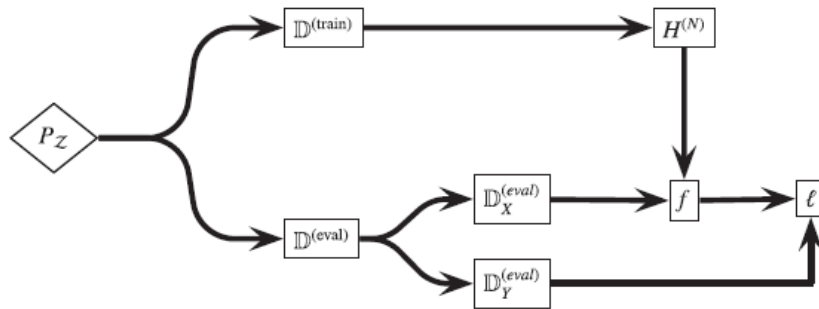
- Depicts every phase of a learning algorithm except model selection.
- Real-world objects ω from the universe Ω are drawn according to a distribution P_Ω .
- The measurement map ξ parses each ω into a data point $x \in \mathcal{X}$.
- The feature selector FS applies a (possibly data-dependent) mapping φ to build $\mathcal{D}^{(\text{train})}$ and $\mathcal{D}^{(\text{eval})}$.
- The learning algorithm $H^{(N)}$ uses $\mathcal{D}^{(\text{train})}$ to select a hypothesis f .
- The predictions of f on $\mathcal{D}^{(\text{eval})}$ are assessed by the loss function ℓ .
- Unlike panel (b), which assumes data drawn directly from $P_{\mathcal{Z}}$, panel (a) makes the data-collection phase explicit.

5. Define inductive bias in the context of learning algorithms. Present some examples.

- Definition: a set of (implicit) assumptions used to create generalizations from a set of observations. Examples of such assumptions:
 - Ockham’s razor: prefer the simplest hypothesis consistent with the observations.
 - Naive Bayes: assume maximal conditional independence among features given the label.
 - Support vector machines: prefer the hypothesis with maximal margin between classes.
- Needed because the book only considers inductive learning: generalizing from prior experience to unseen instances. In contrast with:
 - Deductive learning, which derives new facts from an existing set of axioms or background knowledge, rather than generalizing a pattern from examples.
 - Transductive learning, which predicts labels only for a specific, already-given set of test points, without ever committing to a general hypothesis over the whole input space.
 - Analogical (instance-based) reasoning, which predicts for a new instance by directly comparing it to stored past cases, again without forming an explicit general rule.
- It is what lets the learner prefer one generalization over another, even though the observed data alone would be consistent with many.



(a) The complete learning framework.



(b) The learning framework with implicit data collection.

Figure 2.1 Diagrams depicting the flow of information through different phases of learning.
(a) All major phases of the learning algorithm except for model selection. Here objects from the space Ω are drawn from the distribution P_Ω and parsed into measurements that then are used in the feature selector FS . It selects a feature mapping ϕ that is used to create training and evaluation datasets, $\mathbb{D}^{(train)}$ and $\mathbb{D}^{(eval)}$. The learning algorithm $H^{(N)}$ selects a hypothesis f based on the training data, and its predictions are assessed on $\mathbb{D}^{(eval)}$ according to the loss function ℓ .
(b) The training and prediction phases of learning with implicit data collection phases. Here the data are assumed to be drawn directly from P_Z instead of being drawn from P_Ω and subsequently mapped into the space Z by the measurement process.

6. What is empirical risk? Why can a machine learning algorithm be described as an empirical risk minimization procedure?

- Empirical risk: the average loss of a hypothesis f over the observed training sample,

$$\tilde{R}_N(f) = \frac{1}{N} \sum_{(x,y) \in D} \ell(y, f(x)), \quad N = |D|.$$

- Computed directly from the finite, observed dataset D .
- Contrasts with the true risk, which would require averaging over the whole (unknown) distribution P_Z .
- Since P_Z is unavailable, most statistical learning algorithms minimize $\tilde{R}_N(f)$ instead.
- This is precisely what qualifies them as empirical risk minimization procedures.

7. Explain the sentence: “Minimizing the average loss on the training data is often used as a surrogate for minimizing the expected loss on unseen data.”

- The learner’s actual goal is to minimize the risk $R(P_Z, f)$, the expected loss on new data.
- That quantity cannot be computed because P_Z is unknown.
- What can be computed is the average loss on the sample already in hand, the empirical risk $\tilde{R}_N(f)$.

- Minimizing $\tilde{R}_N(f)$ is used as a surrogate for minimizing $R(P_{\mathcal{Z}}, f)$.
- Justification: under the stationarity assumption and appropriate regularity conditions, generalization bounds (Vapnik, 1995) relate the two quantities.

8. What is the stationarity assumption?

- The assumption that training data and evaluation data are drawn from the same distribution $P_{\mathcal{Z}}$, as in Figure 2.1.
- Justifies using the empirical risk as a surrogate for the true risk.
- Without it, low error on the training sample gives no guarantee about performance on new data.
- The book later studies scenarios, largely arising from adversarial behavior, that deliberately violate this assumption.

9. Define each of the symbols below: ω , Ω , x , $\mathcal{X}$, D , x_i , $\mathcal{X}_i$.

ω : An individual real-world object or event (e.g., a single email or network packet), before measurement.

Ω : The space, or universe, of all such real-world objects.

x : The measured, composite data point resulting from applying ξ to some $\omega \in \Omega$; $x \in \mathcal{X}$.

$\mathcal{X}$: The space of all possible data points, $\mathcal{X} \triangleq \mathcal{X}_1 \times \mathcal{X}_2 \times \cdots \times \mathcal{X}_D$.

D : Most commonly, the dataset itself: the indexed set of N data points $\{x^{(i)}\}_{i=1}^N$ obtained by applying ξ to N real-world objects.

Depending on typeface, D (italic) is also used for the dimensionality of a data point, i.e., the number of measurements composing $x = (x_1, \dots, x_D)$.

x_i : The i -th individual measurement, or feature, of ω ; the i -th coordinate of x .

$\mathcal{X}_i$: The space from which the i -th feature x_i is drawn (e.g., real-valued, boolean, categorical).

10. How does the book define data collection?

- The application of the measurement map ξ to a sequence of N real-world objects $\omega^{(1)}, \omega^{(2)}, \dots, \omega^{(N)}$.
- Produces an indexed set of N data points $\{x^{(i)}\}_{i=1}^N \subseteq \mathcal{X}^N$.
- This indexed set is what the book calls a dataset, denoted D .
- Represents the sequence of observations the learner uses to generalize to future events.

11. What is the common assumption made by all the learning algorithms studied in the book regarding the structure of the data?

- Data points are independently and identically distributed (i.i.d.).
- Sampled from an unknown but stationary distribution.
- Individual methods depend on this assumption to different degrees.

12. What is the difference between explanatory variables and response variables? How do these concepts relate to the data point x , the label y , and the paired space $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$?

- Observations are partitioned into two sets.
- Explanatory variables (covariates, inputs, predictor variables): the quantities that are observed.
- Response variables (output, outcome variables): the unobserved quantities to be predicted.
- In the classification setting, explanatory variables correspond to the data point $x \in \mathcal{X}$.
- The response variable corresponds to the label $y \in \mathcal{Y}$.
- A datum is the pair (x, y) , belonging to the Cartesian product $\mathcal{Z} \triangleq \mathcal{X} \times \mathcal{Y}$.
- Instances are assumed drawn from a joint distribution $P_{\mathcal{Z}}$ over that paired space.

13. What is feature selection, and how does it differ from the measurement process ξ ? Why does the book choose not to distinguish between the two?

- Feature selection: a mapping $\varphi_D : \mathcal{X} \rightarrow \hat{\mathcal{X}}$ that transforms measured data into a space of task-relevant features.
 - Measurement ξ : fixed, data-independent, and typically an irreversible physical process.
 - Feature selection is usually data-dependent, chosen based on the whole dataset D , and can be redone by reprocessing the original measurements.
 - The book merges the two steps because the attacks it studies target downstream aspects of learning, not the measurement or feature-extraction pipeline.
 - As a result, $\mathcal{X}$ is written in place of $\hat{\mathcal{X}}$ throughout.
14. What is a hypothesis space $\mathcal{F}$? Illustrate with an example how assumptions about the form of the hypotheses can restrict this space.
- $\mathcal{F}$: the family of all candidate functions $f : \mathcal{X} \rightarrow \mathcal{Y}$ the learning algorithm may select from.
 - Most general case: the set of all such functions, $\mathcal{F} \triangleq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$.
 - Example restriction: with $\mathcal{X} = \mathbb{R}^D$, restrict to generalized linear functions $f_{a,b}^\beta(x) \triangleq \beta(a^\top x + b)$.
 - $\beta(x) = \mathbb{I}[x > 0]$ with $\mathcal{Y} = \{0, 1\}$: yields the family of all halfspaces on $\mathbb{R}^D$.
 - β the identity with $\mathcal{Y} = \mathbb{R}$: yields the family of linear functions on $\mathbb{R}^D$.
 - Both are much smaller than the unrestricted space of all functions from $\mathcal{X}$ to $\mathcal{Y}$.
15. What is the difference between a learning model and a training procedure? Use the location estimation example (mean versus median) to explain why this distinction is useful for understanding a learner's vulnerabilities.
- Model: captures the assumptions made about the observed data; restricts the hypothesis space and encodes prior knowledge or inductive bias.
 - The model does not specify how a particular hypothesis is chosen, that is the role of the training procedure.
 - Example model: $X \sim \mathcal{N}(\theta, 1)$, data drawn independently from a unit-variance Gaussian centered at an unknown θ .
 - The sample mean and the sample median are two different training procedures valid under this same model.
 - Distinguishing model from procedure matters for security: identical models can hide very different vulnerability profiles.
 - The mean is highly sensitive to even a single corrupted point.
 - The median is comparatively robust to such corruption.
16. Describe the supervised learning framework. How is a learning algorithm $H^{(N)}$ formally defined, and what does it mean for such an algorithm to have a randomized element?
- Observed data: a paired dataset $D = \{(x^{(i)}, y^{(i)})\}$, drawn from an unknown distribution $P_{\mathcal{X} \times \mathcal{Y}}$.
 - Goal: select the classifier $\hat{f} \in \mathcal{F}$ with the smallest expected prediction cost.
 - Formal definition: $H^{(N)} : \mathcal{Z}^N \rightarrow \mathcal{F}$, a mapping from N training examples to a hypothesis.
 - Randomized element: the output additionally depends on a random variable $R \sim P_{\mathbb{R}}$ (e.g., random initialization or subsampling).
 - Written explicitly as $H^{(N)} : \mathcal{Z}^N \times \mathbb{R} \rightarrow \mathcal{F}$.
 - The resulting hypothesis f then depends on both the training data D and the realization of R .
17. What is the difference between batch learning and online learning?
- Batch learning: the learner trains once on the full $D^{(\text{train})}$, producing $\hat{f}$, then is evaluated on $D^{(\text{eval})}$, with no retraining in between.
 - Online learning: a repeated process where the learner continually retrains on evaluation data as labels become available.
 - In pure online learning, prediction and retraining occur at every new data point.
 - Batch learning can be seen as a single epoch of the more general online process.

18. In a binary classification setting applied to security problems, what do the positive and negative classes typically represent? How does the anomaly detection setting studied in Chapter 6 differ from standard binary classification?

- Positive class ($y = +$): malicious or intrusive instances.
- Negative class ($y = -$): benign, normal instances.
- Anomaly detection (Chapter 6) still makes the same two predictions: normal versus anomalous.
- Key difference: the training set typically contains only negative (normal) instances.
- No labeled positive (anomalous) instances are available at training time.
- The detector is instead calibrated to a desired false-positive rate on held-out normal data.

19. What is the difference between the risk $R(P_{\mathcal{Z}}, f)$ and the empirical risk $\tilde{R}_N(f)$? Why does the learner minimize the latter instead of the former?

- Risk:

$$R(P_{\mathcal{Z}}, f) \triangleq \int_{(x,y) \in \mathcal{Z}} \ell(y, f(x)) dP_{\mathcal{Z}}(x, y),$$

the true expected loss of f over the entire (unknown) distribution $P_{\mathcal{Z}}$.

- Empirical risk:

$$\tilde{R}_N(f) = \frac{1}{N} \sum_{(x,y) \in D} \ell(y, f(x)),$$

the average loss of f over the finite observed sample $D \sim P_{\mathcal{Z}}$.

- The learner minimizes $\tilde{R}_N(f)$ because $P_{\mathcal{Z}}$ is unknown, making $R(P_{\mathcal{Z}}, f)$ impossible to compute directly.
- $\tilde{R}_N(f)$ is computable from the available data and, under stationarity, serves as a usable estimate of $R(P_{\mathcal{Z}}, f)$.

20. What is regularization, and what problem does it address? In Equation 2.1, what do $\rho(f)$ and λ represent?

- Addresses overfitting: when $\mathcal{F}$ is too expressive, a hypothesis can fit the training sample exactly (e.g., a lookup table) yet fail to generalize.
- Regularization keeps a large hypothesis space, but penalizes the complexity of the chosen hypothesis.
- The learner selects $\hat{f}$ to minimize the modified objective

$$\tilde{R}_N(f) + \lambda \cdot \rho(f).$$

- $\rho : \mathcal{F} \rightarrow \mathbb{R}$: a measure of the complexity of a hypothesis.
- $\lambda \in \mathbb{R}^+$: the regularization parameter, controlling the tradeoff between empirical risk and hypothesis complexity.

21. What are false positives (FP) and false negatives (FN)? What are the false-positive rate (FPR) and the false-negative rate (FNR), and why is choosing a tradeoff between them an application-specific decision?

- False positive (FP): a normal (negative) instance mistakenly labeled positive.
- False negative (FN): a malicious (positive) instance mistakenly labeled negative.
- FPR / FNR: the frequencies at which the predictor makes each of these two kinds of mistakes.
- There is no single objectively correct point on the FPR/FNR tradeoff.
- Spam filtering: blocking a legitimate email (FP) may be considered worse than letting spam through (FN).
- Intrusion detection: missing an actual attack (FN) is often far costlier than an extra false alarm (FP).
- The appropriate operating point is therefore inherently an application-specific decision.

22. Why does the book restrict its analysis primarily to binary classification, and which other learning paradigms are mentioned but left out of the discussion?

- Many core security applications reduce naturally to two classes: harmful versus benign.

- Examples: host and network intrusion detection, virus and worm detection, spam filtering.
- Binary classification (together with anomaly detection, Chapter 6) is enough to cover these applications while keeping the notation and threat analysis tractable.
- Mentioned but not pursued: semi-supervised learning, unsupervised learning, reinforcement learning.
- These are noted as not directly bearing on the book's discussion.